ClearDate — Data Sources Reference

Where to find each value in `drugs.json`, and how to read it. Links were checked October 2026.

Field-by-field source map

Field	Primary source	Cross-check
halfLifeHours	FDA label (DailyMed), section 12.3 Pharmacokinetics; EU/UK SmPC section 5.2 for non-US drugs	Baselt; Goodman & Gilman PK appendix; primary PK papers (PubMed)
apparentHalfLifeHours (depots, ER)	FDA label for the specific depot/ER product, section 12.3	Product monograph; pivotal PK study cited in the label
metabolites, formedFrom, fraction	FDA label section 12.3 (Metabolism); Baselt	DrugBank metabolism section; ClinPGx pathways; primary mass-balance studies
molecularWeight	PubChem compound page	DrugBank
cyp2d6.substrate, halfLifeMultiplier	FDA label (PM vs EM exposure); CPIC guideline where one exists	ClinPGx drug label annotations; Flockhart table
cyp2d6.inhibitor (strength)	FDA Table of Substrates, Inhibitors and Inducers; FDA examples for healthcare professionals	Flockhart table
cyp2d6.inhibitionType	Primary literature on time-dependent / mechanism-based inhibition	FDA label (drug interaction studies, section 7 and 12.3)
qt.tier	CredibleMeds (KR / PR / CR categories) — enter by hand, cite, do not redistribute their list	FDA label sections 5 (Warnings) and 12.2; thorough QT study results
qt.weight	Your own clinical judgement, informed by hERG IC50 and clinical QTc data	IUPHAR / ChEMBL (hERG activity); primary literature
receptors	IUPHAR/BPS Guide to Pharmacology	PDSP Ki Database; ChEMBL; DrugBank targets
lingeringEffects	Primary literature and clinical guidelines (e.g. benzodiazepine withdrawal guidance)	Your own clinical experience; note it in source
typicalDoseMg (prescribed drugs)	FDA label section 2 (Dosage and Administration)	SmPC section 4.2
aliases	FDA label (brand names); Drugs@FDA	EUDA drug profiles and NPS Discovery for street names

Pharmacokinetics and labels

Source	What it gives you	Link
DailyMed (NLM)	Current FDA labels. Section 12.3 has half-life, metabolism and PM/EM data.	https://dailymed.nlm.nih.gov
Drugs@FDA	Approval packages and older label versions; clinical pharmacology reviews.	https://www.accessdata.fda.gov/scripts/cder/daf/

Source	What it gives you	Link
UK eMC (SmPCs)	Labels for drugs not approved in the US (e.g. domperidone, moclobemide, zopiclone).	https://www.medicines.org.uk/emc
EMA	EU product information (SmPC section 5.2).	https://www.ema.europa.eu
Baselt, *Disposition of Toxic Drugs and Chemicals in Man*	The single best reference for half-lives and metabolites of illicit, novel and older drugs. Book (Biomedical Publications); use the latest edition.	Print / library
Goodman & Gilman's *Pharmacological Basis of Therapeutics*	PK data appendix for common prescription drugs.	Print / AccessMedicine
PubChem	Molecular weights, structures, identifiers, linked literature.	https://pubchem.ncbi.nlm.nih.gov
DrugBank	Aggregated PK, metabolism and targets. Good for orientation; confirm against primary sources.	https://go.drugbank.com
PubMed	Primary PK, mass-balance and QT studies.	https://pubmed.ncbi.nlm.nih.gov

CYP2D6 and pharmacogenomics

Source	What it gives you	Link
FDA Table of Substrates, Inhibitors and Inducers	Index drugs and strong / moderate / weak inhibitor classifications used in drug development.	fda.gov — DDI table
FDA examples for healthcare professionals	Clinically relevant CYP substrates, inhibitors and inducers.	fda.gov — CYP examples
Flockhart Table (Indiana University)	Quick lookup of substrates, inhibitors and inducers by isoform.	drug-interactions.medicine.iu.edu
ClinPGx (formerly PharmGKB)	Curated pharmacogenomic annotations, drug labels and pathways.	https://www.clinpgx.org
CPIC	Guidelines with phenotype definitions and exposure changes for CYP2D6 substrates.	https://cpicpgx.org
PharmVar	CYP2D6 allele definitions (for interpreting a patient's genotype report).	https://www.pharmvar.org

QT and cardiac risk

Source	What it gives you	Link
CredibleMeds	Known / possible / conditional risk of TdP. Free registration. Their terms restrict redistribution, so enter tiers by hand and cite.	https://crediblemeds.org
FDA labels	Section 5 warnings, section 12.2 thorough QT study results (mean and upper-bound ΔΔQTc).	https://dailymed.nlm.nih.gov

Source	What it gives you	Link
ChEMBL	hERG activity data, useful for setting relative qt.weight.	https://www.ebi.ac.uk/chembl/

Receptor targets

Source	What it gives you	Link
IUPHAR/BPS Guide to Pharmacology	Curated targets with action type (agonist, antagonist, etc.) and affinities.	https://www.guidetopharmacology.org
PDSP Ki Database (UNC)	Large set of measured Ki values; strong coverage of psychedelics and psychiatric drugs.	https://pdsp.unc.edu
ChEMBL	Bioactivity data across targets.	https://www.ebi.ac.uk/chembl/
DrugBank	Target lists with action labels; verify against IUPHAR or primary data.	https://go.drugbank.com

Illicit and novel psychoactive substances

Source	What it gives you	Link
Baselt	Half-lives and metabolites for most illicit drugs and many novel ones.	Print / library
EUDA (formerly EMCDDA) drug profiles	Pharmacology, metabolism, street names, adulterant trends.	https://www.euda.europa.eu
NPS Discovery (CFSRE)	Monographs and alerts on novel opioids (nitazenes), benzodiazepines and stimulants, incl. metabolites.	https://www.npsdiscovery.org
PubMed	Case reports and PK studies for designer drugs.	https://pubmed.ncbi.nlm.nih.gov

Ibogaine-specific and enzyme-recovery literature

Look these up on PubMed and confirm the full citations before relying on them.

- Glue P, et al. Ascending-dose study of noribogaine in healthy volunteers: pharmacokinetics, pharmacodynamics, safety, and tolerability. *J Clin Pharmacol*. 2015. — noribogaine PK and concentration–QTc relationship.
- Glue P, et al. Influence of CYP2D6 activity on the pharmacokinetics and pharmacodynamics of a single 20 mg dose of ibogaine in healthy volunteers. *J Clin Pharmacol*. 2015. — CYP2D6 effect on ibogaine exposure.
- Mash DC, et al. Ibogaine detoxification transitions opioid and cocaine abusers between dependence and abstinence: clinical observations and treatment outcomes. *Front Pharmacol*. 2018. — ibogaine/noribogaine in treatment settings; see also Mash's earlier ibogaine PK papers cited there.
- Knuijver T, et al. Safety of ibogaine administration in detoxification of opioid-dependent individuals: a descriptive open-label observational study. *Addiction*. 2022. — clinical QTc course after ibogaine.
- Koenig X, Hilber K. The anti-addiction drug ibogaine and the heart: a delicate relation. *Molecules*. 2015. — hERG data for ibogaine and noribogaine.
- Global Ibogaine Therapy Alliance. Clinical Guidelines for Ibogaine-Assisted Detoxification. — screening and washout practice.

- O'Mathúna B, Farré M, et al. The consequences of 3,4-methylenedioxymethamphetamine induced CYP2D6 inhibition in humans. *J Clin Psychopharmacol*. 2008; and Yubero-Lahoz S, et al. on recovery of CYP2D6 activity after MDMA. — enzyme recovery time after mechanism-based inhibition.
- Yang J, et al. Cytochrome P450 turnover: regulation of synthesis and degradation, methods for determining rates, and implications for the prediction of drug interactions. *Curr Drug Metab*. 2008. — CYP2D6 degradation half-life (default for the recovery setting).

Reading the sources consistently

- **Half-life:** use the terminal elimination half-life in healthy adults after a single dose unless the label gives steady-state values that differ meaningfully (e.g. methadone, fluoxetine); record which one in `sourceNotes`.
- **Ranges:** use the reported range or mean ± 2 SD for min/max. If only a mean is published, set min and max to the mean and note it.
- **Depots and extended release:** enter the release-limited apparent half-life from the product label, not the drug's intrinsic half-life.
- **CYP2D6 multipliers:** derive from the ratio of half-life (or AUC) in poor versus normal metabolisers in the label or CPIC guideline. Leave at 1 if no data.
- **fraction (formedFrom):** often not published precisely. Use mass-balance or urinary recovery data where available; otherwise enter a best estimate and say so.
- **Molecular weight:** use the free base unless doses are conventionally expressed as the salt; note which.
- **Verified:** tick it only after you have checked every numeric field against a primary source.